

Question 1

- a. The crabs dataset is a data frame with 200 observations. This data describes 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus* and *variegatus* collected at Fremantle, W. Australia. The data contains 8 variables; species (factor with 2 levels), sex (factor with 2 levels), Index (integers), frontal lobe (mm), rear width (mm), carapace length (mm), carapace width (mm) and body depth (mm).

- b. R code

```
library(MASS)
data(crabs)
str(crabs)
```

Output

```
'data.frame':  200 obs. of  8 variables:
 $ sp      : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1
1 1 1 ...
 $ sex     : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2
2 2 2 ...
 $ index: int   1 2 3 4 5 6 7 8 9 10 ...
```

- c. R code

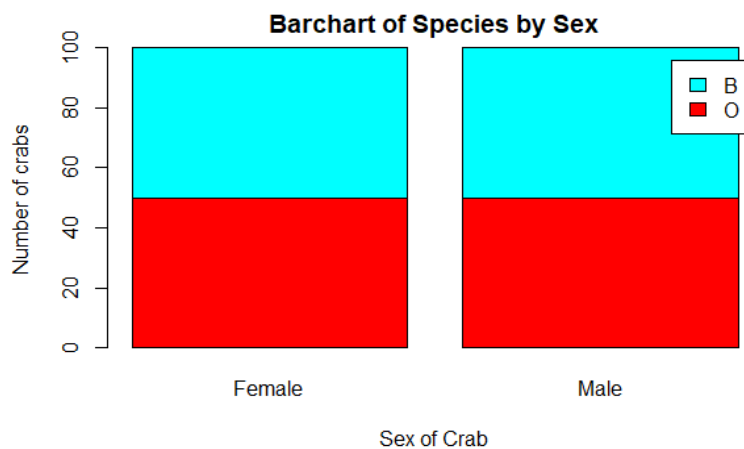
```
SpeciesSex_table <- table(crabs$sp,crabs$sex)
rownames(SpeciesSex_table) <- c('Blue','Orange')
colnames(SpeciesSex_table) <- c('Female',
'Male')
print(SpeciesSex_table)
```

Output

	Female	Male
Blue	50	50
Orange	50	50

d. R code

```
par(mar = c(4, 4, 2, 2))
barplot(SpeciesSex_table, main = 'Barchart of Species by Sex',
        ylab = 'Number of crabs', xlab = 'Sex of Crab', col =
        rainbow(2), legend = (c('O', 'B')))
```

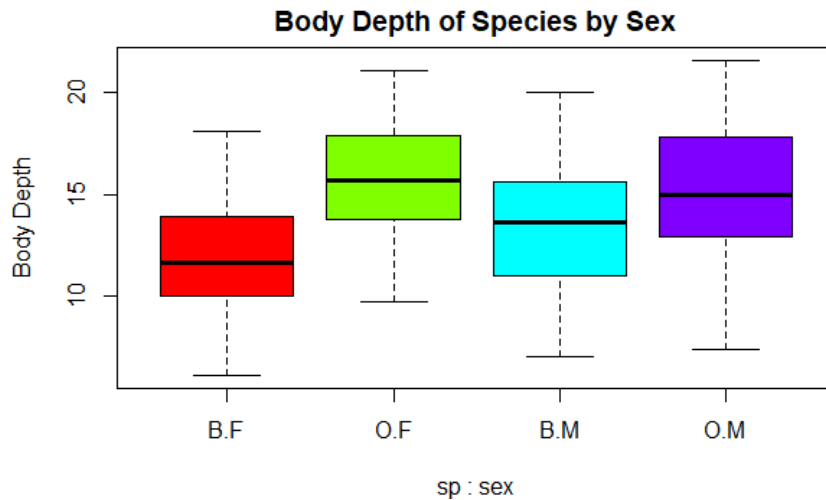


- e. The dataset is balanced across all groups. From the frequency table, there is equal crabs of the same sex and species. This is further backed by the graph as there is equal share of sex across the species.

Question 2

a. R code

```
boxplot(BD~ sp+sex, data= crabs, main='Body Depth of Species by
Sex', ylab = 'Body Depth', 'xlab = Species by Sex', col =
rainbow(4))
```



- b. Blue female crabs have the lowest body depth while orange male crabs have the highest body depth among the 4 groups. Female crabs have the lowest variability in their body depth while males have a higher variability of body depth. While all four groups have no outlier, orange males is skewed to the left meaning that there are a lot of crabs with high body depths.

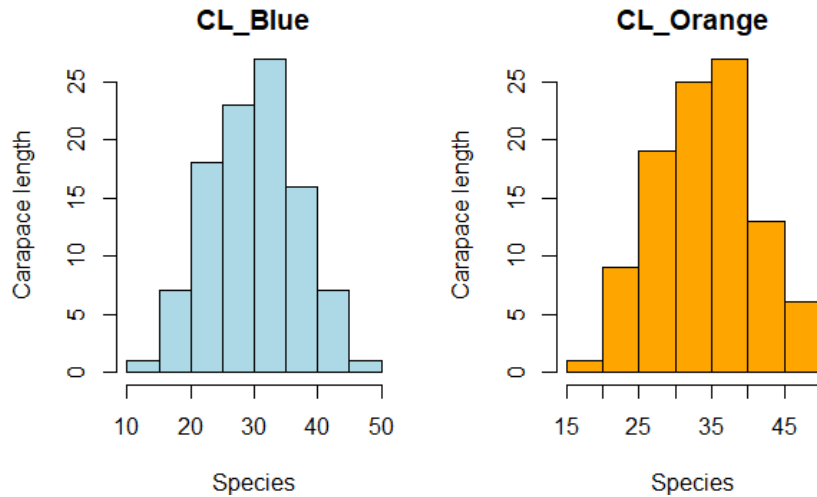
Question 3

```
par(mfrow=c(1,2))

hist(crabs$CL[crabs$sp == 'B'], main = 'CL_Blue', ylab = 'Carapace length', xlab = 'Species', col = 'lightblue')

hist(crabs$CL[crabs$sp == 'O'], main = 'CL_Orange', ylab = 'Carapace length', xlab = 'Species', col = 'orange')

par(mfrow = c(1,1))
```



- b. From the above graph, both CL for blue and orange crabs has approximately a normal distribution. Blue crabs have a lower average carapace length of about 30mm as compared to orange crabs which have an average carapace length of about 35mm

Question 4

- a. R codes

```
meansAll <- aggregate(crabs[, 4:8],
                      by = list(species = crabs$sp,
                                sex = crabs$sex),
                      FUN = mean, na.rm = TRUE)
print(meansAll)
```

Output

	species	sex	FL	RW	CL	CW	BD
1	B	F	13.270	12.138	28.102	32.624	11.816
2	O	F	17.594	14.836	34.618	39.036	15.632
3	B	M	14.842	11.718	32.014	36.810	13.350

- b. R codes

```

group_labels <- with(meansAll, paste(species, sex, sep =
"."))

mat <- t(as.matrix(meansAll[, c("FL", "RW", "CL", "CW",
"BD")]))

colnames(mat) <- group_labels

mat <- t(mat)
mat <- t(mat)

cols <- c(rainbow(4))

par(mar = c(5, 5, 4, 10))

bp <- barplot(mat,

               beside = TRUE,

               col = cols,

               ylim = c(0, max(mat) * 1.15),

               ylab = "Mean (mm)",

               xlab = "Measurement",

               main = "Mean Crab Measurements by Species and
Sex",

               names.arg = colnames(mat),

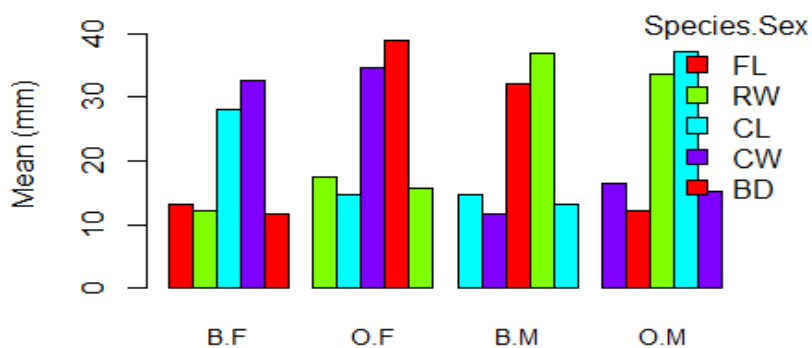
               cex.names = 0.9)

legend("topright", inset = c(-0.12, 0),

       legend = rownames(mat), fill = cols, bty = "n", title

```

Mean Crab Measurements by Species and Sex



Question 6

The crabs dataset is a data frame with 200 observations. This data describes 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus* and *variegatus* collected at Fremantle, W. Australia. The data contains 8 variables; species (factor with 2 levels), sex (factor with 2 levels), Index (integers), frontal lobe (mm), rear width (mm), carapace length (mm), carapace width (mm) and body depth (mm). The data set has equal females and males across the 2 species.

Blue female crabs have the lowest body depth while orange male crabs have the highest body depth among the 4 groups. Female crabs have the lowest variability in their body depth while males have a higher variability of body depth. While all four groups have no outlier, orange males is skewed to the left meaning that there are a lot of crabs with high body depths.

Both CL for blue and orange crabs has approximately a normal distribution. Blue crabs have a lower average carapace length of about 30mm as compared to orange crabs which have an average carapace length of about 35mm.